

Automatic File Organizer Using Python

Project Summary / Note

Objective: The main objective of this project is to automatically organize files into separate folders based on their file types. This reduces manual work and keeps files properly arranged.

Technology Used

- Programming Language: **Python**
- Modules: **os** and **shutil**

How the Project Works

1. The program selects a source folder named **Test_Files**.
2. It reads all files present inside the folder.
3. It checks the extension of each file, such as .jpg, .png, .pdf, .mp3, and .mp4.
4. It identifies the destination category using a Python dictionary.
5. It automatically creates the required folder if it does not already exist.
6. It moves the file into the corresponding folder using the **shutil** module.

File Organization

File Extension	Destination Folder
.jpg	Images
.png	Images
.pdf	Documents
.mp3	Music
.mp4	Videos

Important Python Functions Used

os.listdir() – Reads the files and folders inside the source directory.

os.path.isfile() – Checks whether an item is a file.

os.path.splitext() – Extracts the file extension.

os.makedirs() – Creates destination folders automatically.

shutil.move() – Moves files from one location to another.

Example of Organization

Before: Test_Files contains photo.jpg, notes.pdf, song.mp3, and movie.mp4.

After running the program:

- Images → photo.jpg

- Documents → notes.pdf
- Music → song.mp3
- Videos → movie.mp4

Advantages

- Saves time and reduces manual effort.
- Keeps files organized automatically.
- Easy to understand and modify.
- Automatically creates required folders.
- Useful for managing large numbers of files.

Conclusion

The Automatic File Organizer is a simple Python automation project that organizes files according to their extensions. It demonstrates the practical use of Python's os and shutil modules for file and folder management.